
Lab 8

Question 1

Write client-server code (both single and multi-server code using sockets and multi-threading) such that:

- The server sends a string to the client
- The client receives it and prints it

Question 2

Write client-server code (both single and multi-server code using sockets and multi-threading) such that:

- The server sends a string to the client
- The client receives it and prints it, replies back with another string
- The server receives the other string and print it

Question 3

Write a client-multi server code using sockets and multi-threading where you use the client to chat with a dummy "math" server. The protocol between the client and server is as follows.

- The client asks the user for input. The user enters a simple arithmetic expression string (e.g., "1+2", "5 - 6", "3*4"). The user's input is sent to the server via the connected socket.
- The server reads the user's input from the client socket, evaluates the expression, and sends the result back to the client.
- The client should display the server's reply to the user, and prompt the user for the next input, until the user terminates the client program with "Ctrl C".

Your server should behave like any real server. It should be able to handle several clients concurrently. It should work fine as clients come and go. Your server should always keep running (until terminated with "Ctrl C"), and should not quit for any other reason.

At the very minimum, the server should be able to handle addition, multiplication, subtraction, and division operations on two integer operands. So, some sample test cases are:

- User types: 1 + 2, server replies 3
- User types: 2 * 3, server replies 6
- User types: 4 - 7, server replies -3
- User types: 30 / 10, server replies 3